CMAAC: Combining Multiattention and Asymmetric Convolution Global Learning Framework for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification methods based on deep learning (DL) techniques have succeeded wildly. However, the high-dimensional nonlinearity, spectral mixing, and difficulty in labeling training samples of HSI still hinder the accuracy of HSI classification. Several patch-free-based methods were proposed for HSI classification and have attracted attention. Nevertheless, recent patch-free methods have focused on low- or high-order interactions, ignoring the abundant middle-order interactions, failing to capture the multiorder interactions in the context of HSI, and difficulty extracting discriminative features when the sample data are imbalanced. In this article, a combining multiattention and asymmetric convolution (CMAAC) global learning framework was proposed for insufficient utilization of spectral–spatial information and imbalanced HSI samples. In CMAAC, the channel convolutional long short-term mechanism and multiorder spatial aggregation block (CLMS), which aim to capture multiorder interactions information and extract more discriminant spectral–spatial features effectively, is proposed. The pyramid-enhanced attention mechanism (PEAM) alleviates the information loss in feature flow, retains more spatial information, and better solves the challenge of scale diversity in different land-cover types. The asymmetric convolutional structure at the end of the model highlights the influence of local key feature points and improves its representation ability. Additionally, the framework introduces a joint loss function (JLF) to reduce misclassified classes caused by the imbalance between well-classified and hard-classified samples. We conducted experiments on four benchmark datasets and compared them with the state-of-the-art approaches, demonstrating the model's effectiveness and superiority.**

***Index Terms*— Deep learning (DL), fully convolutional network (FCN), hyperspectral image (HSI) classification, patch-free.**

I. Introduction

HYPERSPECTRAL remote sensing (HRS) image is a 3-D data cube acquired by a corresponding sensor that captures dozens or hundreds of contiguous narrow spectral bands [1], [2]. Due to the abundant spectral–spatial features [3], hyperspectral image (HSI) has a wide range of applications in many fields, such as geological exploration, urban planning, military target identification, forestry resource monitoring, agricultural development, and environmental protection [4], [5], [6], [7], where the HSI classification is a common task and critical procedure, aiming to predict the class label for each pixel in HSI data.

In the early HSI classification work, researchers focused primarily on exploiting spectral features, including support vector machine (SVM) [8], maximum likelihood classification (MLC) [9], multinomial logistic regression (MLR) [10], and random forest (RF) [11]. Dimensionality reduction is a commonly used method to solve the problem of excessive bands in HSI, such as principal component analysis (PCA) [12], linear discriminant analysis [13], and other spectral dimensionality reduction methods. These traditional methods only extract the shallow feature representations of HSI and fail to learn the abstract features. Meanwhile, pixel-level classifiers disregard spatial context, and adjacent pixels can provide vital spatial information for improving the performance of HSI classification. Hence, researchers have been increasingly interested in classification methods that merge spectral–spatial features, including Gabor filters [14], extended morphological profiles [15], and multiple kernel learning [16]. However, these approaches heavily depend on hand-crafted feature descriptors. At present, deep learning (DL) technology has good performance in image classification [17], [18] and is widely used in HSI classification, with commonly used algorithms, including stacked autoencoder (SAE) [19], deep belief networks (DBNs) [20], and convolutional neural networks (CNNs) [21]. SAE and DBN can automatically learn the features of data through the network and have better classification effects, but the requirement for a large number of parameters, complex training, and 1-D vector input limits their ability to effectively consider spatial information. CNN demonstrates remarkable performance in HSI classification, and its improved, fusion algorithms based on CNN have gradually become mainstream methods. Compared with traditional methods, CNN-based methods significantly improve classification accuracy.

The HSI classification methods of DL are classified into three categories based on the input format: pixel-level [19], patch-based [22], and patch-free [23]. Using only the spectral

Manuscript received 6 November 2023; revised 10 December 2023 and 23 January 2024; accepted 30 January 2024. Date of publication 2 February 2024; date of current version 16 February 2024. This work was supported by the Wuhan Science and Technology Plan Project-Application Basic Frontier Project under Grant 2019010701011403. *(Corresponding author: Xubing Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3361555

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sequence of each pixel as input features can improve classification performance, but it is not always satisfactory. In cases where two different objects have the same spectrum, they need to be identified by their shapes and textures. Therefore, the importance of spatial features in enhancing HSI feature representation and classification accuracy cannot be overlooked. The CNN-based methods attempt to explore the spectral–spatial features from the adjacent pixels for HSI classification, adjacent pixels in the square box from an HSI patch. Numerous variants of CNNs have made significant progress in patch-based local learning frameworks, e.g., 3-D-CNN [22], multichannel CNN [24], recurrent neural networks (RNNs) [25], and other novel models, such as capsule neural networks [26], multistage self-guided separation network [27], representation-enhanced status replay network [28], multistage information complementary fusion network [29], and so on. To alleviate the dilemma of the insufficient of labeled samples faced in HSI classification, methods based on unsupervised learning [30], semi-supervised learning [31], self-supervised learning [32], and small samples learning [33] are also gradually proposed. However, the image patches generated by adjacent pixels overlap with each other, which leads to computational redundancy and affects the speed of the patch-based learning framework, and only capturing the local spatial features within the patch cannot obtain long-range spatial dependence. To address the above issues, a spectral–spatial fully convolutional network (SSFCN) [23] was proposed to effectively avoid the limitations of patch classification methods and enable pixel-by-pixel classification labeling, which is a double-branch network to extract spectral–spatial features separately, and the dense conditional random field (CRF) is introduced into the framework to further balance the local and global information. Furthermore, the FreeNet framework [34] was proposed for HSI classification, which maximizes the extraction of global spatial contextual information for the whole image using a spectral attention-based encoder and a lightweight decoder structure, achieves high precision on public datasets, and reduces redundant computation. Moreover, a fully contextual network (FullyContNets) [35] was presented, which equips the scale attention module (SAM) and two fully contextual modules. This integration enables the model to effectively capture contextual information across positions, channels, and features simultaneously.

Recently, a patch-free learning framework has been increasingly used in HSI classification. Although the extraction of global context and spatial relationships effectively improves classification accuracy, the flow of global information is accompanied by the loss of local information and detail, while challenges remain in how to utilize spectral–spatial features adequately. In this article, we propose a patch-free HSI classification framework that combines the channel convolutional long short-term mechanism and multiorder spatial aggregation block (CLMS), pyramid-enhanced attention mechanism (PEAM), and asymmetric convolution structure (ACS) to effectively learn representative features, restoring clear category boundaries and feature details. The CLMS combines channel convolutional long short-term mechanism (CLSTM) and multiorder spatial aggregation block (MSAB) to extract spectral–spatial features. One branch is used to extract long short-term correlations of spectral features, and the other is used to capture multiorder interactions in context. The fusion of the two branches can effectively deal with complex spectral–spatial relations and extract more discriminant features. The PEAM mechanism uses parallel convolution with different dilation rates to expand the receptive field while preserving local detail features. It can effectively extract fine-grained multiscale spatial information and achieve adaptive aggregation of the multiscale contextual information, better solving the scale difference problem. ACS enhance CNN's convolution kernel skeleton and recovery of image edge information. The framework employs skip connections between the spatial feature in the encoder and decoder can provide high-resolution detail information. Moreover, a joint loss function (JLF) is introduced to address the class imbalance problem.

The main contributions of this article are as follows.

1) A combining multiattention and asymmetric convolution (CMAAC) global learning framework was proposed to input the whole image to finish the HSI classification task. This framework can more effectively utilize and focus on the spectral–spatial features of HSI, solve the problem of sample imbalance, and learn from global spatial information to get better accuracy.
2) To explore more discriminative spectral–spatial features, the CLMS is proposed. The CLSTM module models the long-term dependence of spectral features, and the MSAB module effectively captures multiorder interactions to increase the diversity of features. PEAM module is developed to extract fine-grained multiscale spatial information to better address the challenge of scale diversity in different land-cover types.
3) The skip connection enhances connectivity across different layers to propagate the gradients better. Meanwhile, an asymmetric convolution structure is used at the end to strengthen the influence of local edge information, and deep strip convolution can complement grid convolution, facilitating the extraction of strip features and recovery of strip target details.
4) To solve the datasets with class imbalance, a novel JLF was designed to assist with training the proposed method, CMAAC. Using the proposed JLF to reduce the weight of well-classified samples and focus on hard-classified samples can effectively promote the convergence speed of the model.

II. Related Work

DL methods have gradually become one of the most critical techniques in HSI classification, which aims to assign a label to each pixel. Early methods based on spectral–spatial DL attempt to incorporate spatial information by flattening operations to supplement spectral information. Hu et al. [36] proposed a 1-D-CNN to classify HSI in a supervised manner. Mou et al. [37] first applied RNN to HSI classification and demonstrated its effectiveness in HSI classification. Hao and Prasad [38] used 1-D-CNN combined with RNN to capture

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contextual information from the spectral bands used for HSI classification. Zhan et al. [39] proposed a novel generative adversarial network (GAN) to alleviate the problem of insufficiently labeled HSI pixels. The above methods mainly use the flattening operation to exploit the spatial information and spread the HSI cube into dimensional vectors, in which the spatial structure information has been corrupted. Recently, patch-based and patch-free classification methods have been successfully applied to HSI classification and have proven effective in dealing with the high-dimensional and complex HSI. Nonetheless, researchers continue to explore new methods to improve the performance of HSI classification tasks.

A. DL-Based HSI Classifiers Under the Patch-Based Learning Framework

The patch-based framework aims to classify the central pixel of the target image patch, and numerous CNN-based networks have been developed to mine the spectral–spatial information of HSI. Niu et al. [40] used 2-D-CNN to perform HSI classification taking the neighborhood window around each labeled pixel as a training sample. Zhao and Du [41] proposed a spectral–spatial feature-based classification framework (SSFC), which joined PCA dimensionality reduction and DL techniques for spectral and spatial feature extraction and finally stacked the features together for classification. Hamida et al. [22] developed a 3-D-CNN, which used a 3-D patch with a small input size to process spectral–spatial information and improve classification performance. Roy et al. [21] designed the hybrid-CNN model to enhance classification performance by combining 3-D-CNN and 2-D-CNN. Subsequently, a spectral–spatial residual network (SSRN) was proposed in [42], which can make the model deeper and extract abundant spatial and spectral features to improve classification performance. Zhang et al. [43] designed a diverse region-based CNN to learn contextual features from six different HSI patches. Zhang et al. [24] proposed a dual-channel CNN to extract spectral–spatial features separately. The dual-branch multiattention mechanism (DBMA) network and a 3-D channel and spatial attention-based multiscale spatial–spectral residual network (CSMS-SSRN) were proposed to enhance the representation of the HSI features [31], [44]. Liu et al. [2] proposed a central attention network (CAN), the scaled dot-product central attention (SDPCA) tailored for HSI aims to extract spectra–spatial information by collecting the information of a query pixel, and its neighbors that are similar to the query pixel, providing a new perspective for the excellent HSI features extraction method. In order to extract the spectral dependence among adjacent bands, Wu and Prasad [25] introduced an RNN to simulate the dependence between different spectral bands and to learn more discriminative feature representations for HSI classification. Furthermore, Hu et al. [45] considered the 2-D extension structure of the long short-term memory (LSTM) to build a spectral–spatial ConvLSTM 2-D neural network (SSCL2DNN) to model the long-range dependence in the spectral domain.

In recent years, numerous variants of CNNs have been applied to HSI classification. Paoletti et al. [26] proposed a spectral–spatial capsule network to reduce the complexity of the convolutional networks. The deformable HSI classification network (DHCNet) method uses an adaptive spatial context modeling method to capture the complex spatial context in HSI and improve its performance [46]. In addition, Zhan et al. [39] proposed hyperspectral GAN (HSGAN), which was the first time used the GAN for the HSI semi-supervised classification. Zhong et al. [47] introduced a spectral–spatial GAN and CRF (SS-GAN-CRF) classification method based on the GAN and the probability graph model. Graph convolutional networks (GCNs) are able to model long-range spatial relations between samples in HSI. In the multiscale dynamic GCN (MDGCN), the original HSI is divided into multiple superpixel blocks, and a dynamic graph structure is formed according to the multilayer neighborhood of each superpixel, which obtains a better feature representation and reduces the complexity of the model training [48]. With the introduction of the vision transformer and its variants, transformer-based networks have been gradually applied to the HSI classification [49], [50], [51]. Hong et al. [52] rethought HSI classification from a sequence perspective and proposed a highly flexible backbone network, SpectralFormer, which better utilized and represented the spectral–spatial features. Xu et al. [53] designed a multiscale and cross-level attention learning (MCAL) network, which combines convolution with multihead self-attention (MHSA) to extract multiscale local features and global pixel features.

Meanwhile, solving the small sample classification problem also achieves excellent classification results. Baassou et al. [54] developed a spectral–spatial information synthesis algorithm for HSI image classification, which uses spatial pixel association (SPA) by exploiting spectral information divergence (SID), and spectral clustering to reduce region numbers and improve classification accuracy. Gao et al. [33] adopted a depth model based on induction networks to improve the classification performance of HSI under few-shot conditions. Pan et al. [55] proposed a small-scale data-based method named multigrained network (MugNet), a lightweight 3-D-CNN for semi-supervised HSI classification. Fang et al. [5] designed a lightweight 3-D-CNN and generated pseudolabels by approximating rank-order clustering to fully use large unlabeled HSI samples. Zhang et al. [56] introduced the structural optimization transmission network (SOT-Net) for collaborative land cover classification of HSI and light detection and ranging (LiDAR) data, which can produce satisfactory classification results even for small-size training samples. A single-source domain expansion network (SDEnet) for cross-scene HSI classification based on domain generalization ensures the reliability and effectiveness of domain extension, and achieves good experimental results [57]. Guan and Lam [58] investigated a cross-domain contrastive learning (CDCL) framework to learn representations of HSI in an unsupervised mode. Zhang et al. [59] proposed a topological structure and semantic information transfer network (TSTnet) for cross-scene HSI classification, the domain alignment from both statistical distribution and topology perspectives, and the process of dynamically updating the graph to realize

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more accurate feature embedding and graph construction. Hu et al. [60] proposed a dual-adjustment mode-based cross-domain meta-learning (DMCM) method for few-shot HSI classification, and the class-covariance metric (CCM) is employed to optimize decision boundary and alleviate the misclassification problem. Mei et al. [30] proposed a 3-D convolutional autoencoder (3-D-CAE) for feature extraction in an unsupervised mode. The network is trained without the labeled training samples, which maximizes obtaining the most discriminative spectral and spatial information.

The patch size determines the network's local receptive field and affects the extraction of the spatial background information. Meanwhile, the cropped HSI patch's edges inevitably contain interfering pixels with different labels from those of the current HSI patches, and these interfering pixels will affect the extraction of spectral–spatial features. Although supervised classification addresses the issue of inadequate and imbalanced samples, its incorrect split patches will increase the amount of computation and eventually lead to classification failure.

B. DL-Based HSI Classifiers Under the Patch-Free Learning Framework

The patch-free framework utilizes the entire HSI as an input and the sparse training samples as the labels. The training and inference processes of the fully convolutional network (FCN) are performed through intensive feed-forward computation and backpropagation in one complete pass, thereby eliminating any limitations on image inputs and significantly improving the inference efficiency. U-Net [61], a representative FCN, captures multiscale information through the bottom–up paths, top–down paths, and lateral connections. SegNet [62] utilizes an FCN-based encoder–decoder structure to recover the spatial resolution and obtain more accurate segmentation results. HSI classification involves assigning a unique label to each pixel, similar to semantic segmentation. However, due to the limitation of available labeled samples, the given sample contains only a discrete set of labeled pixels rather than a group of labeled images. To address this issue, Xu et al. [23] proposed a spectral–spatial FCN for HSI classification with an end-to-end, pixel-to-pixel structure, avoiding patch extraction. A new mask matrix and a spectral–spatial weighting factor were developed to assist backpropagation and adaptively learning the features during the training phase. Zheng et al. [34] proposed a fast patch-free global learning framework for HSI classification, which models the long-range spatial to maximize the use of global spatial features. Zhu et al. [63] developed a spectral–spatial-dependent global learning framework, proposed a global convolutional long short-term memory (GCL) module to distinguish similar land covers effectively, and designed a global joint attention mechanism (GJAM) to reweight and model the extracted features. Jiang et al. [64] proposed a fully convolutional spatial propagation network (FCSPN), which consists of a 3-D-FCN and a convolutional spatial propagation network (CSPN) to effectively capture spatial and spectral information. Wang et al. [35] propose a fully contextual networks for hyperspectral scene parsing, which combines multiscale features and self-attention mechanisms that can effectively capture the more generic nonlocal contexts. Tu et al. [65] proposed a nonlocal-dependent learning FCN, which utilizes context-aware modules and GCL neural network for learning deep features and enhancing the long-range dependencies of spectral and spatial dimensions at different phases to perform better classification.

Although patch-free HSI classification methods based on FCN have demonstrated promising results, several important issues remain to consider. How to capture multiorder interaction information in context? How to extract more discriminative spectral–spatial features of HSI? How to solve the problem of HSI sample imbalance? There are still challenges to achieve high-accuracy classification results. Therefore, we propose the CMAAC framework for HSI classification.

III. Proposed Method

To fully utilize the spectral–spatial information, obtain abundant multiorder features, and extract more detailed and complete target contours, the CMAAC framework is proposed for HSI classification. This framework is based on an encoder–decoder FCN, which takes the entire image as an input, and the overall architecture is shown in Fig. 1. A parallel module CLMS is implemented to extract the discriminative spectral–spatial features, in which the long-term dependence of the spectral domain is modeled by combining attention with ConvLSTM and a specially designed aggregation module to acquire multiorder features. The PEAM module alleviates the information loss in the feature flow and enhances the representation capacity of targets at different scales. By adding the asymmetric convolution at the end, the model could capture the local key points features, recover the fine-grained details at edges, and improve the generalization performance. A JLF is introduced to relieve the class imbalance problem effectively. A novel skip connection is used to merge semantic information of different resolutions. In Sections III-A–III-D, we will provide the details of the CLMS, PEAM, ACS, and JLF.

A. Channel Convolutional Long Short-Term Mechanism and Multiorder Spatial Aggregation Block

As shown in Fig. 2, the CLMS module comprises the CLSTM and MSAB. The CLSTM and MSAB are concatenated through channels to obtain the final feature aggregation, and the CLMS can exploit the local and global spatial dependencies to enable the model to represent the complex spectral–spatial relationships better and nonlinearly transform the input features into more discriminative spectral–spatial features.

1) The CLSTM is an essential part of the encoder layer, as shown in Fig. 3. To effectively distinguish the similar spectral features of the HSIs, the input feature map is divided into n groups according to the number of the channels. Also, n is the time step length of ConvLSTM to capture the interdependence of long-term spectral features. The weights of the spectral information are adjusted by the channel attention mechanism, which is beneficial to distinguish the features with similar spectral characteristics.

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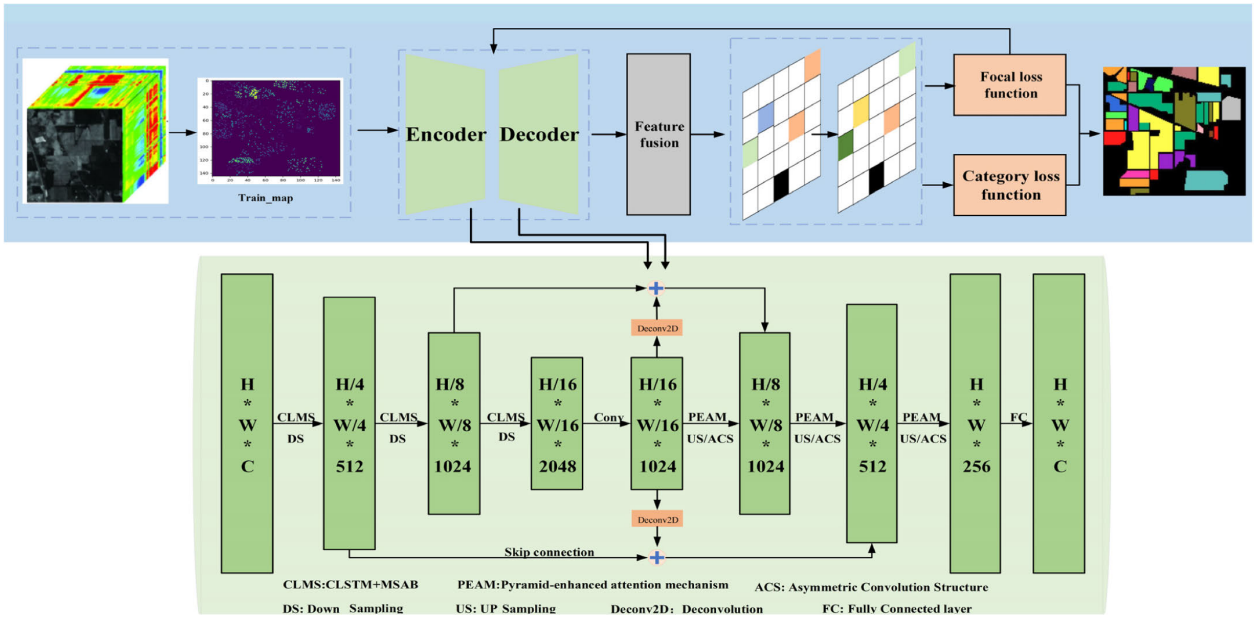


Fig. 1. Overview of the CMAAC global learning framework. The CMAAC framework includes two core components: the encoder–decoder-based FCN and lateral connections.

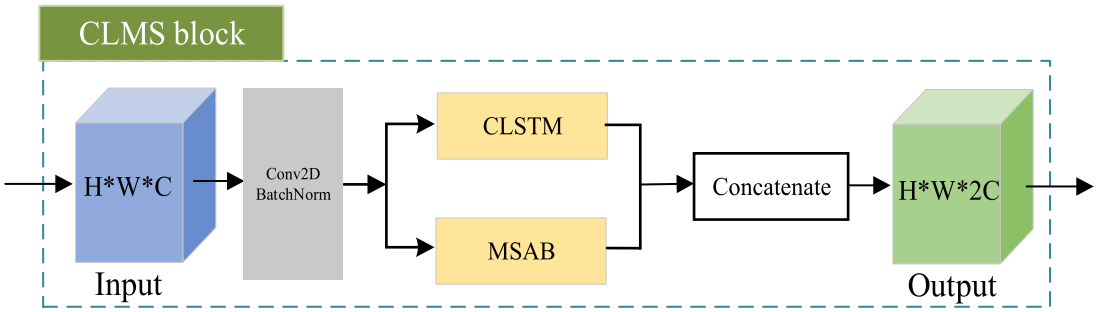


Fig. 2. Schematic of the CLMS.

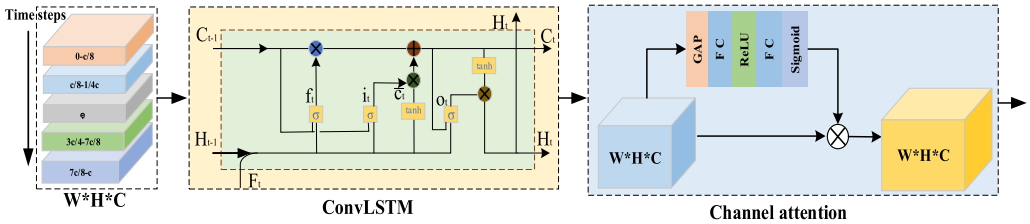


Fig. 3. Schematic of the CLSTM.

a) The ConvLSTM network consists of several cells, and the ConvLSTMCells include four parts: the forget gate f_t, the input gate i_t, the output gate o_t, and the storage cell c_t. The structure diagram is shown in Fig. 3. f_t, i_t, and o_t act as three different control gates for clearing, receiving, updating the storage cells, and completing the data transmission and processing. The calculation formula of the cell can be written as

$$\begin{aligned} i_t &= \sigma(W_{xi} * x_t + W_{hi} * h_{t-1} + W_{ci} \circ c_{t-1}) \\ f_t &= \sigma\left(W_{xf} * x_t + W_{hf} * h_{t-1} + W_{cf} \circ c_{t-1}\right) \\ \bar{c}_t &= \tanh(W_{xc} * x_t + W_{hc} * h_{t-1}) \\ c_t &= f_t \circ c_{t-1} + i_t \circ \bar{c}_t \\ o_t &= \sigma(W_{x0} * x_t + W_{ho} * h_{t-1} + W_{co} \circ c_t) \\ h_t &= o_t \circ \tanh(c_t) \end{aligned} \tag{1}$$

where t represents the tth time step, x_t represents the input of the current cell, and c_{t-1} and h_{t-1} represent the state and output of the last cell, respectively. $*$ means the convolutional operation, W denotes the 2-D convolution filter with a $k \times k$ kernel, and k is the size of the convolution kernel, respectively. W_{ci}, W_{cf}, and W_{co} are the peephole weight, W_{xi}, W_{xf}, W_{xo}, and W_{xc} are the gated unit convolution kernels for x_t, W_{hi}, W_{hf}, W_{ho}, and W_{hc} are the gated unit convolution kernels for h_{t-1}, $\circ$ represents the Hadamard product, and σ is the sigmoid activation function.

b) The channel attention mechanism enhances the interdependence between spectral features and improves the quality of the features generated by ConvLSTM to improve performance. As shown in Fig. 3, the input feature map is passed through the average pooling layer and compressed into a feature vector [1, 1, c], the correlation between channels is

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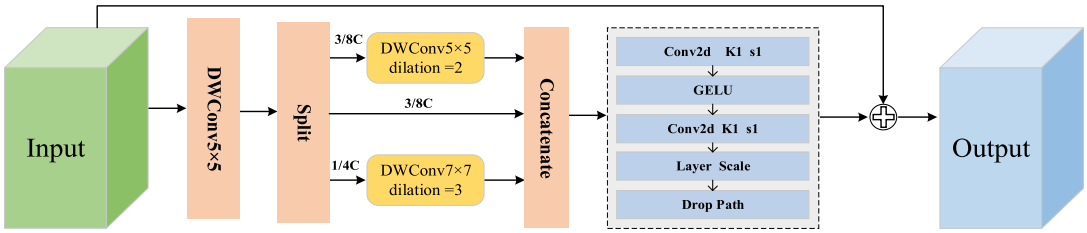


Fig. 4. Schematic of the MSAB.

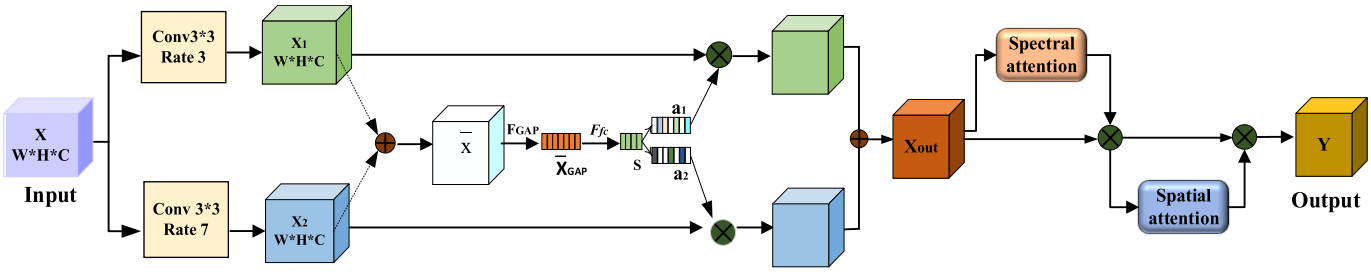


Fig. 5. Schematic of the PEAM.

constructed through a fully connected layer, and a weight value is generated for each feature channel. Finally, the normalized weights are multiplied by the original input feature map channels to generate the weighted feature map.

The channel attention mechanism can express it

$$Mc(\boldsymbol{X}) = \sigma(FC1(\delta(FCo(AvgPool(\boldsymbol{X})))) \times \boldsymbol{X} + \boldsymbol{X} \quad (2)$$

where $\boldsymbol{X}$ represents the input feature, σ denotes the sigmoid function, FC represents the fully connected layer, $M_c(\boldsymbol{X})$ represents the output, and δ represents the ReLU function.

2) The MSAB structure diagram is shown in Fig. 4. The size of the convolutional kernel limits the receptive field of the CLSTM module, and multiple convolutional layers need to be stacked to expand the receptive field and discover global features. Adding more convolutional layers to ConvLSTM would increase the risk of overfitting the model, and increasing the number of parameters can lead to the ineffectiveness of the model. If only the spatial information of the current time is entered and the past spatial information is ignored, it would be difficult to explore the spatial dependence in time and only capture the local spatial dependence. Current CNNs focus on low- or high-order interactions and ignore abundant middle-order interactions. The MSAB module contributes to solving the long-range spatial dependence problem while effectively modeling middle-order information across spatial and channel spaces.

Given an input feature $\boldsymbol{X} \in R^{C_\times H\times W}$, we employ three different depthwise convolutions (DWConv) with dilation ratios $d \in \{1, 2, 3\}$ to encode multiorder features in the branch of MSAB. First, apply $\mathrm{DW}_{5\times5,a=1}$ to get the low-order features, $\mathrm{DW}_{5\times5,a=1}$ represents a DWCond layer with a 5×5 kernel, and dilation rate is set to 1; factorized the low-order features $\boldsymbol{X}_1 \in R^{C_{1\times H\times W}}$, $\boldsymbol{X}_2 \in R^{C_{2\times H\times W}}$, and $\boldsymbol{X}_3 \in R^{C_3\times H\times W}$ along the channel dimension, C represents the number of channels, where $C1 + C2 + C3 = C$. Then assign $\boldsymbol{X}_1$ and $\boldsymbol{X}_3$ to $\mathrm{DW}_{5\times5,a=2}$ and $\mathrm{DW}_{7\times7,\alpha=3}$, respectively, through three DWConX layers parallel obtained low, middle, and higher order interactions, and finally, the output of $\boldsymbol{X}_1$, $\boldsymbol{X}_2$, and $\boldsymbol{X}_3$ is concatenated to form multiorder contexts. After that, channel scaling is performed by 1×1 convolution, using the Gaussian error linear unit (GELU) activation function for nonlinear transformation, the layer scale, and the drop path to improve the performance and stability of the model, and finally fused with $\boldsymbol{X}$ to get the final result.

B. Pyramid-Enhanced Attention Mechanism

As the network structure deepens, continuous downsampling causes the loss of image information and spatial relationships. Therefore, a PEAM module is designed, as shown in Fig. 5. Convolution kernels with various receptive fields are constructed using dilated convolutions with different sampling rates. A multiscale pyramid structure is employed to gradually enhance receptive fields and preserve more spatial information by integrating feature maps. This results in clearer and more complete target contours in high-level features. Due to the edge effect, when the sampling rate increases, the convolution kernel's effective weights decrease, leading to a degradation in effectiveness. To address the problems and integrate global context information into the module, we adopt the global average pooling (GAP) used to obtain the image's global features. The next major process to guide the network adaptively generates the weights of different scale features and acquires multiscale contextual information to mitigate the scale differences of the targets. Moreover, to further increase the feature distinction while maintaining the details, a spectral–spatial attention mechanism is utilized to strengthen important channel features and spatial regions, effectively extract fine-grained multiscale information, and improve the expression capability of semantic classification.

The multiscale attention mechanism uses two atrous convolutions with rates of 3 and 7 to extract multiscale features $\boldsymbol{X}_1 \in R^{C_\times\times H\times W}$ and $\boldsymbol{X}_2 \in R^{C_\times\times H\times W}$ from the input feature map $\boldsymbol{X} \in R^{C_\times\times H\times W}$. The features at different scales are element

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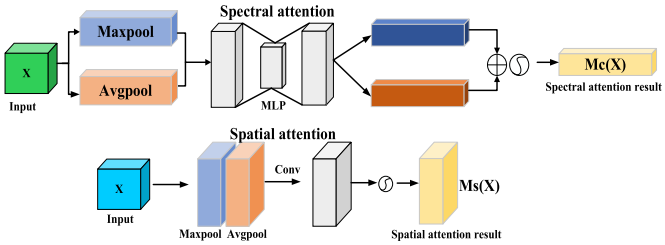


Fig. 6. Schematic of the spectral–spatial attention.

merged to GAP to identify the global spatial information and output the channelwise statistics feature $\overline{X}_{\mathbf{GAP}} \in R^{C\times 1\times 1}$, which aggregates the global information of C channels in the feature at different scales; this part operates as follows:

$$\overline{X}_{\mathrm{GAP}} = F_{\mathrm{GAP}}\left(\sum_{i=1}^{2}\overline{X}_i\right). \tag{3}$$

The output $\overline{X}_{\mathbf{GAP}}$ is fed into a fully connected layer to determine the size of the weight each channel contributes. To reduce the network parameters, the channel of feature vector $\overline{X}_{\mathbf{GAP}}$ is compressed into d [$d = \max(C/16, 32)$] channels, which is equivalent to a squeeze operation, adaptively selecting features. The specific formula for this operation is as follows:

$$S = Ffc\left(\overline{X}_{\mathrm{GAP}}\right) = \delta\left(F_{\mathrm{BN}}\left(W\left(\overline{X}_{\mathrm{GAP}}\right)\right)\right) \tag{4}$$

where δ denotes the ReLU activation function; $F_{\mathrm{BN}}(\cdot)$ is the batch normalization layer; $W \in R^{d\times C}$ which can compress the channel dimension of the output feature maps to 1/16 of that of the feature map $\overline{X}_{\mathbf{GAP}}$. Then, the softmax function is applied to the feature vector $\boldsymbol{S} \in R^{d\times 1}$ to output the channel weights at different scales. It is expressed by the following formula:

$$\mathrm{a}_i = \frac{e^{A_i S}}{\sum_{i=1}^{2} e^{A_i S}} \tag{5}$$

where a_{i} represents the weight of multi-scale $\boldsymbol{X}_{\mathrm{i}}$, $A \in R^{C\times d}$, note that $A_i \in R^{1\times d}$ is the i-th row of A; From the above formula, it can be concluded that the weight of the j-th channel has the following relationship:

$$\sum_{i=1}^{2} \mathrm{a}_i\left[\mathrm{j}\right] = 1. \tag{6}$$

After calculating the weights of the features at different scales, $\boldsymbol{X}_{\mathbf{out}}$ of the final scale features is obtained

$$\boldsymbol{X}\mathbf{out} = \sum_{\mathrm{i}}^{2}(\mathrm{a}_i \times \boldsymbol{X}_i). \tag{7}$$

The spectral–spatial attention is shown in Fig. 6, where GAX and GAP denote the global maximum pooling and global average operations, respectively, with f and σ denoting the convolution operation and activation function. The spectral–spatial attention to the information within the network flows efficiently and obtains the output $\boldsymbol{Y}$

$$Mc(\boldsymbol{X}) = \sigma(\mathrm{MLP}(\mathrm{GAP}(X)) + \mathrm{MLP}(\mathrm{GAX}(\boldsymbol{X})) \tag{8}$$

$$Ms(X) = \sigma\left(f^{7\times 7}(\mathrm{GAP}(\boldsymbol{X}); \mathrm{GAX}(\boldsymbol{X}))\right) \tag{9}$$

$$Y = MS((MC(\boldsymbol{X}\mathbf{out}))). \tag{10}$$

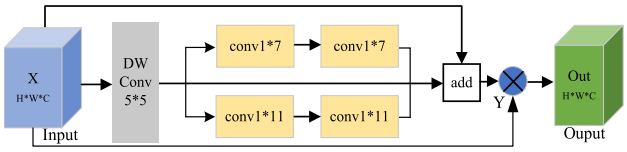


Fig. 7. Flowchart of the asymmetric convolution structure.

C. Asymmetric Convolution Structure

The asymmetric convolution structure at the end helps enhance the effect of local key points, and the asymmetric convolution in two-depth directions reinforces the squared convolution kernels in horizontal and vertical directions and complements the grid convolution, which contributes to extract the strip features and recover the strip target details. As shown in Fig. 7, the 1×7 and 7×1 convolutions are utilized instead of the 7×7 convolution, which can reduce the number of parameters. Additionally, the outputs of different layers are combined to improve the learning quality and the model's robustness. The process can be expressed as follows:

$$\boldsymbol{Y} = \mathrm{Conv}_{1\times 1}\left(\sum_{\mathrm{i}=0}^{2}\mathrm{L_i}(\mathrm{DWConv}(\boldsymbol{X})) + \mathrm{DWConv}(\boldsymbol{X})\right) \tag{11}$$

$$\mathbf{Out} = \boldsymbol{Y} \times \boldsymbol{X} \tag{12}$$

where $\boldsymbol{X}$ denotes the input feature, $\boldsymbol{Y}$ represents the output of the asymmetric structure, DWConv represents the depthwise convolution, L_i denotes the different branches, and $\boldsymbol{Y}$ and $\boldsymbol{X}$ are multiplied element by element to obtain **Out**.

D. Joint Loss Function

In DL techniques, the loss function plays an essential role as a performance metric for the model. Combining the JLF with the proposed algorithm to deal with class-imbalanced datasets can improve the model convergence speed and maintain the model's classification performance to a certain extent. By reducing the weights of easy-to-classify samples, the model focuses more on hard-to-classify samples during training to achieve higher classification accuracy. The experimental results show that the multicategory-based JLF has good results and helps to alleviate the hard and well-classified sample imbalance phenomenon. It can be expressed by the following formula:

$$\mathrm{FL} = -\sum_{t=1}^{n}(1-p_t)^r y_t{}^* \log(p_t) \tag{13}$$

$$\mathrm{CL} = -\sum_{t=1}^{n} a^* y_t^* \log(p_t) \tag{14}$$

$$\mathrm{JLF} = \mathrm{FL} + \mathrm{CL} \tag{15}$$

where the JLF denotes the value of the multicategory loss function corresponding to a pixel point on the image; $-\log(p_t)$ is the initial cross-entropy function value; n represents the number of categories; y_t and p_t represent the true label and the predicted probability of the tth category, respectively. r is the focus function, taking a value between

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$[0, +\infty]$, the larger r is, the stronger of the JLF to suppress the class imbalance phenomenon, but choosing too large r may affect the overall accuracy (OA) of the model. $a = (\Sigma_{i=1}^{n} \text{num}_t)/\text{num}_t$ is used to control the intercategory balance, and num_t denotes the number of samples belonging to category t. Based on several experiments, r was set to 2 in our experiments.

IV. Experimental Results and Analysis

In this section, four HSIs are selected for experiments to illustrate the effectiveness of the proposed algorithm, including Indian Pines (IP), Salinas (SA), Pavia University (PU), and Houston University (HU), with IP and SA are farmland scenes, while PU and HU are relatively complex urban scenes. The quantitative and qualitative analysis of the proposed model compared with advanced HSI classification methods, including HybridSpectralNet (HybridSN) [21], SSRN [42], DBMA [31], SpectralFormer [52], U-Net [61], FreeNet [34], FullyContNet [35], and SSFCN-CRF [23]. All the experiments were performed based on the PyTorch library on the GeForce RTX 2080 Ti.

A. Experimental Settings

1) Model Parameters: With different resolutions and channels, we modify the upsampling and downsampling strategies for each size accordingly. For upsampling, we compress the number of channels of the feature using a 1×1 convolution layer and increase the resolution using bilinear interpolation. For 1/2 scale downsampling, we use 3×3 convolution to modify the number of channels and the resolution. For the 1/4 scale, we use 4×4 convolution with a step size of 4 to reduce the resolution.

2) Parameter Optimization: The proposed framework was trained with 200 iterations of training processes for IP, SA, and PU and 500 iterations for HU. The initial learning rate is set to 0.01 and multiplied by $(1 - (\text{iteration}/\text{max_iteration}))^{\text{power}}$, with power $= 0.8$. The stochastic gradient descent (SGD) optimizer is adopted to train the model, the initial momentum is set to 0.9, and the weight decay rate is 0.001.

3) Metrics: In our experiments, the OA, average accuracy (AA), kappa coefficient (Kappa), and the accuracy of each class were used to evaluate the performance of different classification methods. To eliminate the bias introduced by the random selection of training samples, the results were presented in the form of mean value $\pm$ standard deviation by repeating the same set of experiments ten times.

B. Experiment 1: PU Dataset

The PU dataset was acquired in the area of the University of Pavia, Northern Italy, using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor, which has 610×340 pixels with a spatial resolution of 1.3 m, 103 spectral bands were retained, and the wavelength range from 430 to 860 nm. After removing the background pixels, 42 776 were reserved, containing nine urban land cover types, such as trees, asphalt, meadows, and so on. The number of samples in the PU dataset is large, and the various types of features are easily distinguishable; 80 samples of each class were selected as training samples and the rest as test samples, as listed in Table I. The false-color composite picture and the corresponding ground-truth map are shown in Fig. 8 as a visual reference.

TABLE I
Numbers of Train and Test Samples for the PU Dataset

Class	Class name	Training	Test	Total
C1	Asphalt	80	6551	6631
C2	Meadows	80	18569	18649
C3	Gravel	80	2019	2099
C4	Trees	80	2984	3064
C5	Metal Sheets	80	1265	1345
C6	Bare Soil	80	4949	5029
C7	Bitumen	80	1250	1330
C8	Bricks	80	3602	3682
C9	Shadow	80	867	947
Total		720	42056	42776

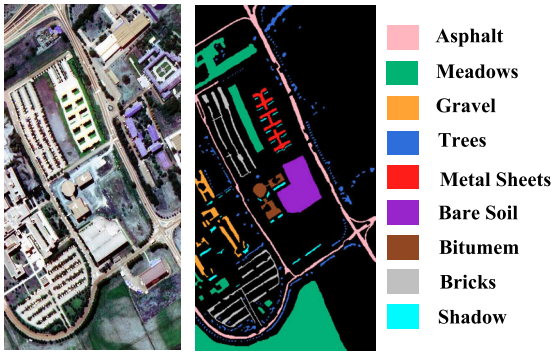


Fig. 8. False-color image and ground-truth map for the PU dataset.

To visually demonstrate the performance of our method, the visual comparison of classification results with different methods is presented in Fig. 9(a)–(i), including HybridSN, SSRN, DBMA, SpectralFormer, U-Net, FreeNet, FullyContNet, SSFCN-CRF, and CMAAC. It can be observed that DBMA and SpectralFormer exhibit poor visual performance and contain a large amount of noise. SSRN has relatively clear profiles but still has a lot of misclassified areas and limited classification performance. Because these methods only utilize the local spatial information of the input 3-D patches, there are many isolated misclassified regions in the classification maps. The black dashed line in the figure outlines part of the highlighted classification detail areas, by comparing Fig. 9(a)–(i), and it is clear that the patch-free method outperforms the patch-based method regarding spatial details, which can be attributed to introduce a more global spatial context. Our method generates classification maps with fewer misclassified pixels, obtains a complete shape of bricks, and accurately distinguishes between meadows and trees. Compared with FreeNet, FullyContNet, and SSFCN-CRF, the proposed method generates a better satisfactory classification map, and the category boundary is closer to the real HSI. The edges of the CMAAC classification map are smoother than other methods, which indicates that the structure can better capture fine-grained details.

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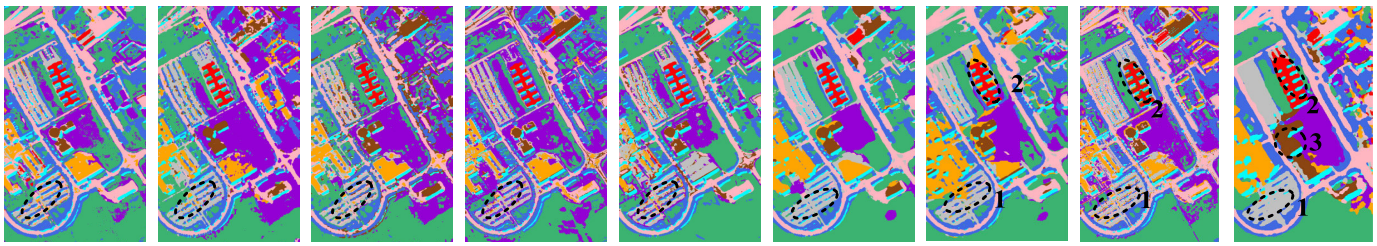


Fig. 9. Visualization of the classification maps for the PU dataset. The black dashed lines outline part of the subtle classification distinctions. (a) HybridSN. (b) SSRN. (c) DBMA. (d) SpectralFormer. (e) U-Net. (f) FreeNet. (g) FullyContNet. (h) SSFCN-CRF. (i) CMAAC.

TABLE II

CLASSIFICATION RESULTS OF HYBRIDSN [21], SSRN [42], DBMA [31], SPECTRALFORMER [52], U-NET [61], FREENET [34], FULLYCONTNET [35], SSFCN-CRF [23], AND CMAAC ON THE PU DATASET

	Patch-batch				Patch-free				
Class	HybridSN	SSRN	DBMA	SpectralFormer	U-Net	FreeNet	FullyContNet	SSFCN-CRF	CMAAC
C1	97.34±0.61	98.29±0.39	89.2±4.09	81.46±0.37	96.94±2.83	98.60±0.68	97.15±1.46	93.83±2.70	**99.13±0.54**
C2	97.67±0.53	98.26±0.43	95.88±2.12	90.78±0.06	86.86±13.55	99.0±1.27	98.92±0.65	97.84±0.78	**99.64±0.40**
C3	89.17±1.67	89.53±3.20	92.75±1.60	95.84±4.64	74.37±14.53	99.85±0.20	97.58±3.07	94.99±2.15	**99.90±0.03**
C4	97.45±0.82	**99.03±0.21**	96.22±0.12	96.95±0.85	98.56±0.52	98.62±0.61	97.89±0.66	98.27±1.22	97.81±0.31
C5	99.97±0.03	**100.0±0.0**	99.94±0.09	99.82±0.25	99.78±0.19	99.98±0.07	99.99±0.02	98.87±1.0	99.78±0.07
C6	94.71±4.15	97.72±2.47	96.87±2.51	95.55±0.24	90.31±7.95	99.86±0.33	99.25±1.46	97.62±1.18	**100.0±0.0**
C7	99.41±0.49	99.33±0.52	98.40±1.39	80.10±0.08	93.93±2.01	99.76±0.43	**99.92±0.22**	95.80±1.56	99.77±0.09
C8	96.52±2.98	94.56±1.96	86.63±0.54	90.53±5.56	85.87±9.27	**99.20±1.05**	98.20±1.40	94.62±2.49	98.94±0.68
C9	**99.92±0.11**	98.20±0.47	99.15±0.57	77.60±4.84	99.86±0.21	99.82±0.31	99.83±0.24	99.91±0.15	97.36±0.80
OA	96.39±1.03	97.09±0.45	94.30±0.25	89.10±0.69	93.93±2.01	99.14±0.57	98.56±0.72	96.81±0.94	**99.31±0.54**
AA	95.19±1.39	96.86±0.61	92.46±0.32	90.72±0.15	85.87±9.27	99.41±0.23	98.72±0.78	95.22±1.54	**99.64±0.40**
Kappa	96.22±1.29	97.42±0.30	94.99±0.18	85.62±0.78	99.86±0.21	98.87±0.75	98.09±0.97	95.78±1.24	**99.90±0.03**

To further evaluate the performance of CMAAC, the OA, AA, Kappa, and per-class precision of compared classification methods are listed in Table II. Our method acquires multiorder information and fully uses the spectral–spatial features to achieve the best classification results. As can be seen from Table II, HybridSN uses 3-D- and 2-D-CNNs to mine more discriminative spectral–spatial information and obtains good classification results. The classification accuracy of SSRN is significantly better than that of HybridSN, DBMA, and SpectralFormer, which indicates that rich spectral–spatial features can be extracted through spectral–spatial residual blocks, and the classification accuracy can be substantially improved. The OA of U-Net is also lower than SSRN, which indicates that the direct application of semantic segmentation advanced networks to HSI classification does not yield better results than advanced patch-based classification methods. Based on the patch-free approach, there are a large number of misclassifications in SSFCN-CRF. FreeNet and FullyContNet use a variety of attention mechanisms to obtain higher results than SSRN. However, they still fail to capture the multiorder interactions that affect HSI classification results. The OA of CMAAC reaches up to 99.31%, which is 1.7%, 0.75%, and 2.5% higher than that of FreeNet, FullyContNet, and SSFCN-CRF, respectively. Except for trees, bricks, and shadows, the proposed CMAAC network achieves a classification accuracy of over 99% on all samples, demonstrating a superior classification performance compared to other algorithms, which verifies that the model has a strong ability to extract discriminative features and can focus on the local key points features of HSI to improve the classification performance significantly. This is attributed to the fact that the CLSTM mechanism is more conducive to extracting abundance information and obtaining discriminative spectral–spatial features. Meanwhile, our PEAM model alleviates the loss of information and retains outstanding features.

C. Experiment 2: IP Dataset

The IP dataset was acquired using the airborne sensor airborne visible/infrared imaging spectrometer (AVIRIS) on a farm test site in Northwest Indiana, USA. The dataset has 145 × 145 pixels with a spatial resolution of 20 m. It includes 16 vegetation types, including corn, oats, and woods, with 200 bands retained ranging from 400 to 2500 nm and 10 249 pixels. A false-color image and corresponding ground truth are shown in Fig. 10. The number of training and test samples are listed in Table III, where alfalfa, grass-pasture-mowed, and oats have only 46, 28, and 20 samples. A highly imbalanced distribution of labeled samples in the IP dataset will lead to training difficulty.

Fig. 11(a)–(i) illustrates the classification results with different methods. The categories with a small number of training samples are difficult to classify, and from the perspective of visual effect, our model has good performance when dealing with imbalanced samples, especially for the isolated target categories of alfalfa, grassing-pasture mowed, and oats. This is due to the introduction of a JLF, which balances the relative loss of easy and hard-to-classify samples. The classification maps obtained by SSRN, DBMA, and SpectralFormer contain a lot of misclassified pixels. A small number of misclassified regions in FreeNet, and FullyContNet,

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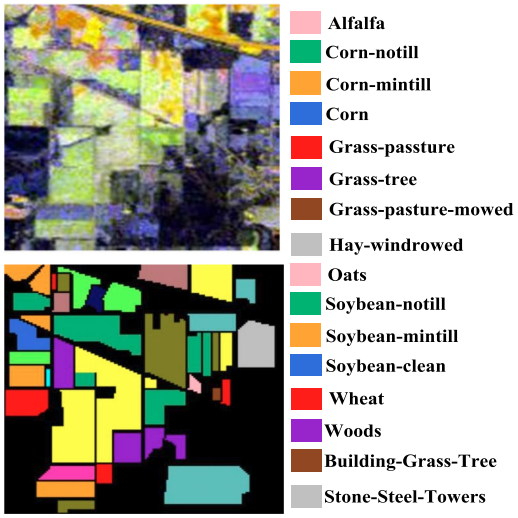


Fig. 10. False-color image and ground-truth map for the IP dataset.

TABLE III
NUMBERS OF TRAIN AND TEST SAMPLES FOR THE IP DATASET

Class	Class name	Training	Test	Total
C1	Alfalfa	26	20	46
C2	Comnotill	80	1348	1428
C3	Commintill	80	750	830
C4	Com	80	157	237
C5	Grass-pasture	80	403	483
C6	Grass-trees	80	650	730
C7	Grass-pasture-mowed	16	12	28
C8	Hay-windrowed	80	398	478
C9	Oats	11	9	20
C10	Soybean-notill	80	892	972
C11	Soybean-mintill	80	2375	2455
C12	Soybean-clean	80	513	593
C13	Wheat	80	125	205
C14	Woods	80	1185	1265
C15	Buildings-Grass-Trees	80	306	386
C16	Stone-Steel-Towers	60	33	93
Total		1073	9176	10249

presumably because these methods perform feature extraction in the entire spatial domain. In addition, the classification maps of FreeNet and FullyContNet show that these methods with better classification performance in categories with similar spectral characteristics, such as corn, grassland, and soybean, and have good generalization ability. However, there is still a handful of noise in the comnotill area, and the soybeans do not have good boundaries. Comparatively, the proposed method has superior performance and strong feature learning capabilities in handling imbalanced datasets.

To verify the model's effectiveness in detail, we present the quantitative indicators for HSI classification in Table IV, where the best accuracy for each row is highlighted in bold. Our proposed method achieved the highest OA, AA, and Kappa, showing a classification advantage in nine categories. Furthermore, our JLF can adjust the sample weights to reduce the weights of easily distinguishable samples, and the PEAM mechanism reduces the loss of features of imbalanced samples, resulting in satisfactory results for imbalanced training samples, such as alfalfa, grass-pasture-mowed, and oats. Therefore, our CMAAC framework effectively alleviates the data imbalance problem.

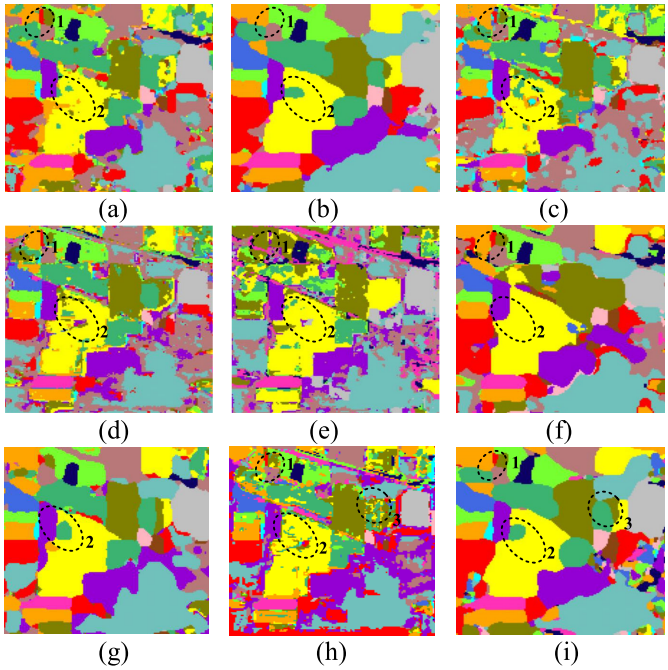


Fig. 11. Visualization of the classification maps for the IP dataset. The black dashed lines outline part of the subtle classification distinctions. (a) HybridSN. (b) SSRN. (c) DBMA. (d) SpectralFormer. (e) U-Net. (f) FreeNet. (g) FullyContNet. (h) SSFCN-CRF. (i) CMAAC.

D. Experiment 3: SA Dataset

To further evaluate the model's effectiveness, we also conducted experiments on the SA dataset, which was collected in the Salinas Valley, CA, USA, and consists of 16 classes, such as celery, fallow, and stubble. The SA dataset contains 512 × 217 pixels in 224 spectral bands, usually removing 20 noise bands affected by water vapor and retaining the remaining 204 spectral bands for experiments with a spatial resolution of 3.7 m per pixel. Fig. 12 shows the false-color composite image and the corresponding ground-truth map, and Table V lists the number of training and test samples.

The classification maps of different methods are shown in Fig. 13. The results demonstrate that Fig. 13(f)–(i) outperforms (a)–(d), and the patch-free method is a better proposal than the patch-based method. Whereas CMMAC and FullyContNet obtain similar accuracy, the CMMAC results contain less noise and achieve better visual performance. The model effectively differentiates between vineyard trellis and celery in the SA dataset with almost no misclassified areas and accurately obtains the soil and fallow boundaries consistent with ground-truth maps. There are obviously interclass boundaries on the HSI with better continuity and significantly less noise in the homogeneous region, which benefit from the CLMS can model long-term spectral dependence and capture multiorder features, and the PEAM mechanism can further enhance feature distinction while retaining spatial details information. In addition, using asymmetric convolution at the end of the model to supplement edge information, make the classification

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TABLE IV

CLASSIFICATION RESULTS OF HYBRIDSN [21], SSRN [42], DBMA [31], SPECTRALFORMER [52], U-NET [61], FREENET [34], FULLYCONTNET [35], SSFCN-CRF [23], AND CMAAC ON THE IP DATASET

	Patch-batch				Patch-free				
Class	HybridSN	SSRN	DBMA	Spectral Former	U-Net	FreeNet	FullyContNet	SSFCN-CRF	CMAAC
C1	**100.0±0.0**	**100.0±0.0**	**100.0±0.0**	98.33±2.36	99.57±0.87	49.5±49.57	**100.0±0.0**	**100.0±0.0**	**100.0±0.0**
C2	88.27±3.91	85.84±3.59	82.34±5.17	85.16±2.85	68.1±15.75	82.14±27.95	94.46±2.71	65.62±2.87	**97.47±1.19**
C3	88.06±2.46	91.79±5.48	90.93±4.12	93.56±0.98	86.28±7.74	93.37±12.31	**98.26±1.26**	85.91±1.83	97.94±1.30
C4	98.26±1.57	99.62±0.27	96.60±3.14	99.36±0.03	99.58±0.68	99.96±0.13	99.93±0.22	90.45±3.92	**99.96±0.13**
C5	96.17±0.77	97.02±1.42	94.29±2.52	93.47±0.42	95.69±3.13	97.28±2.09	**99.14±0.83**	91.40±2.44	99.13±0.74
C6	99.32±0.25	98.95±0.34	96.56±2.22	99.49±0.19	99.08±0.76	99.79±0.21	**99.68±0.30**	92.31±0.88	99.59±0.45
C7	96.60±2.01	98.55±0.72	100.0±0.0	99.23±0.05	92.86±4.79	20.0±40.0	**100.0±0.0**	**100.0±0.0**	**100.0±0.0**
C8	99.85±0.22	99.38±0.06	99.25±2.22	99.78±0.21	100.0±0.0	99.97±0.06	**100.0±0.0**	99.42±0.24	**100.0±0.0**
C9	99.70±0.03	**100.0±0.0**	**100.0±0.0**	**100.0±0.0**	73.5±9.76	20.0±40.0	**100.0±0.0**	**100.0±0.0**	99.5±1.50
C10	87.22±3.03	90.43±5.61	89.57±0.70	91.78±0.61	85.86±4.05	93.19±3.88	97.34±1.39	77.58±4.67	**98.42±1.11**
C11	85.16±4.29	83.88±1.99	81.40±5.57	78.73±6.01	87.37±4.37	83.40±28.30	94.18±2.28	76.32±3.60	**96.68±0.92**
C12	88.31±4.44	97.75±1.16	90.77±2.43	96.56±0.96	88.75±5.79	96.44±7.61	96.88±1.95	77.65±3.91	**98.38±0.79**
C13	99.60±2.86	98.92±0.23	90.84±0.14	**100.0±0.0**	**100.0±0.0**	99.95±0.15	**100.0±0.0**	98.40±0.65	99.95±0.15
C14	96.44±0.91	98.85±1.01	95.22±3.61	89.82±0.86	98.62±0.89	99.23±0.91	99.75±0.43	92.01±1.39	**99.85±0.19**
C15	99.55±0.15	98.84±1.63	99.02±1.13	91.72±2.04	99.17±0.83	**100.0±0.0**	**100.0±0.0**	90.41±2.33	99.97±0.08
C16	**100.0±0.0**	99.83±0.12	97.22±0.21	**100.0±0.0**	**100.0±0.0**	99.90±0.32	99.44±1.67	91.91±9.37	99.89±0.32
OA	92.61±1.12	91.70±2.17	89.09±0.96	88.86±1.83	89.27±2.83	91.17±6.91	96.96±0.55	81.73±1.20	**98.35±0.31**
AA	96.52±1.21	90.50±2.46	87.48±1.06	87.27±2.05	92.46±1.60	81.54±6.90	98.69±0.26	80.06±2.37	**99.17±0.20**
Kappa	91.48±0.48	96.44±1.35	94.75±0.40	94.86±0.63	87.79±3.19	90.07±7.61	96.49±0.64	79.06±1.32	**98.12±0.36**

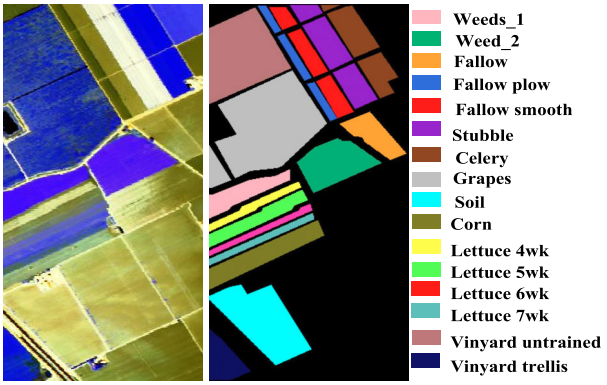


Fig. 12. False-color image and ground-truth map for the SA dataset.

TABLE V

NUMBERS OF TRAIN AND TEST SAMPLES FOR THE SA DATASET

Class	Class name	Training	Test	Total
C1	Brocoli_green_weeds_1	80	1929	2009
C2	Brocoli_green_weeds_2	80	3646	3726
C3	Fallow	80	1896	1976
C4	Fallow_rough_plow	80	1314	1394
C5	Fallow_smooth	80	2598	2678
C6	Stubble	80	3879	3959
C7	Celery	80	3499	3579
C8	Grapes_untrained	80	11191	11271
C9	Soil_vinyard_develop	80	6123	6203
C10	Corn_senesced_green_weeds	80	3198	3278
C11	Lettuce_romaine_4wk	80	988	1068
C12	Lettuce_romaine_5wk	80	1847	1927
C13	Lettuce_romaine_6wk	80	836	916
C14	Lettuce_romaine_7wk	80	990	1070
C15	Vinyard_untrained	80	7188	7268
C16	Vinyard_vertical_trellis	80	1727	1807
Total		1280	52849	54129

map recover precise boundaries, and obtain the complete structure of land covers.

Table VI lists the classification results of different methods on the SA dataset. FreeNet, FullyContNet, and SSFCN-CRF obtain better classification accuracy with OA above 90% and benefit from the global learning framework that fully utilizes global contextual information. Our model achieves the best OA, AA, and Kappa values and shows excellent results in ten categories, with CMAAC obtaining 99.81% of OA and higher than 1.26% compared to the FullyContNet method. Moreover, CMAAC improved the classification accuracy of broccoli_2 by 1.33% and fallow_smooth by 0.37% more than FullyContNet. In addition, CMAAC is more robust in terms of classification accuracy for each class. CMAAC benefits from the shallow layer in the encoder and the deep layer in the decoder to achieve the fusion of semantic information and spatial details, reducing the problem of information loss during the flow of features, and making this method well-suited for the classification of various complex scenes.

E. Experiment 4: HU Dataset

The HU dataset was acquired from the University of Houston, TX, USA, using the CASI-1500 sensor, and contains 349 × 1905 pixels with a spatial resolution of 2.5 m and 144 spectral bands, the wavelength range of 380–1050 nm. It contains 15 categories, such as grass healthy, soil, highway, and so on. Table VII lists the number of training and test samples, and Fig. 14 shows the false-color composite image and the corresponding ground-truth map.

The results of the compared methods are plotted in Fig. 15. As can be seen, the patch-free method achieves good results and directly inputs the entire HSI without any processing and outputs the classification map. The visualization of classification maps generated by different methods is shown in Fig. 16, with (f)–(i) exhibiting more precise and less misclassified regions than other classification methods. One commercial category is easily mistaken for a residential, road, or parking lot due to its similar spectral–spatial features, but these four categories can be better distinguished in the CMAAC.

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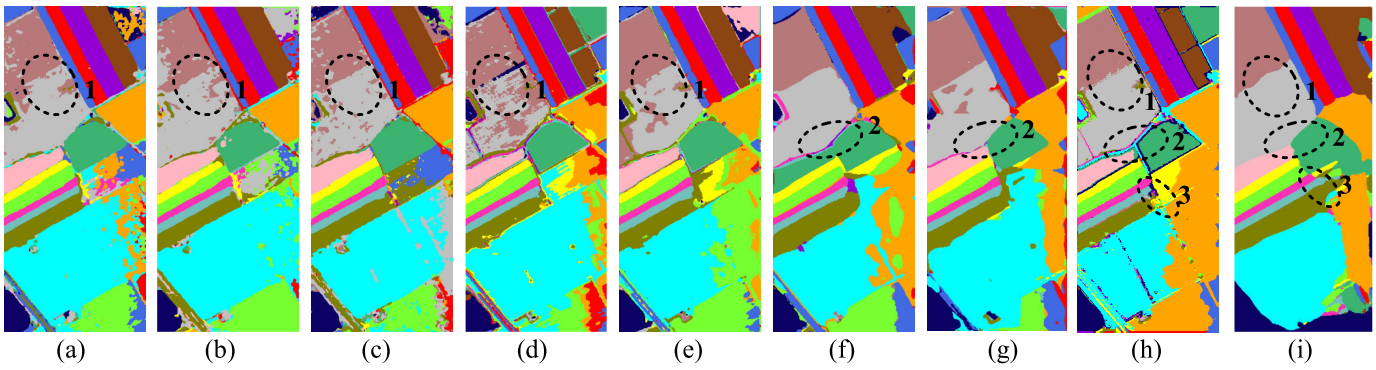


Fig. 13. Visualization of the classification maps for the SA dataset. The black dashed lines outline part of the subtle classification distinctions. (a) HybridSN. (b) SSRN. (c) DBMA. (d) SpectralFormer. (e) U-Net. (f) FreeNet. (g) FullyContNet. (h) SSFCN-CRF. (i) CMAAC.

TABLE VI

CLASSIFICATION RESULTS OF HYBRIDSN [21], SSRN [42], DBMA [31], SPECTRALFORMER [52], U-NET [61], FREENET [34], FULLYCONTNET [35], SSFCN-CRF [23], AND CMAAC ON THE SA DATASET

	Patch-batch				Patch-free				
Class	HybridSN	SSRN	DBMA	Spectral Former	U-Net	FreeNet	FullyContNet	SSFCN-CRF	CMAAC
C1	98.24±0.15	99.46±0.76	99.84±0.16	99.07±0.08	99.98±0.04	79.92±39.96	99.82±0.55	98.20±1.32	**100.0±0.0**
C2	99.79±0.03	99.82±0.17	99.78±0.06	99.95±0.39	99.96±0.04	79.99±39.99	98.66±1.60	99.17±0.73	**99.99±0.02**
C3	99.98±0.02	99.58±0.02	99.94±0.02	89.77±2.55	89.35±16.6	**100.0±0.0**	99.97±0.08	99.92±0.08	**100.0±0.0**
C4	99.89±0.06	99.96±0.04	99.44±0.29	99.85±0.29	99.77±12.3	**99.97±0.07**	99.61±0.30	99.18±0.81	99.94±0.13
C5	98.21±1.33	**99.74±0.18**	99.24±1.04	98.80±0.08	98.63±0.65	99.49±0.51	99.19±0.69	96.75±0.90	99.56±0.29
C6	99.69±0.05	99.92±0.03	99.57±0.40	99.40±0.57	99.95±0.05	**99.98±0.04**	99.92±0.23	99.60±0.11	99.96±0.07
C7	99.97±0.04	99.98±0.01	99.79±0.26	98.66±0.07	99.94±0.05	**99.99±0.04**	99.89±0.19	98.02±1.07	99.96±0.07
C8	89.24±2.48	88.70±1.24	88.22±3.21	58.79±11.54	84.98±8.36	89.25±29.76	97.17±2.82	96.04±1.42	**99.49±0.38**
C9	**99.95±0.06**	98.68±0.04	99.70±0.19	99.83±0.03	99.88±0.10	99.76±0.57	99.82±0.10	98.91±0.65	99.94±0.14
C10	99.03±0.69	99.16±0.33	98.98±0.63	91.31±0.58	83.95±3.76	99.68±0.28	99.13±0.72	90.87±2.39	**99.77±0.49**
C11	99.89±0.07	99.86±0.19	99.76±0.33	99.49±0.04	99.55±0.49	**100.0±0.0**	99.94±0.12	99.19±0.0	**100.0±0.0**
C12	99.62±0.25	99.16±0.26	99.48±0.50	99.67±0.06	99.98±0.02	99.92±0.20	99.74±0.37	99.92±0.11	**99.99±0.03**
C13	**100.0±0.0**	99.72±0.39	**100.0±0.0**	99.28±0.64	99.89±0.21	99.98±0.07	**100.0±0.0**	98.47±0.31	99.99±0.03
C14	99.79±0.12	99.62±0.38	99.36±0.49	99.39±0.07	99.81±0.06	99.91±0.10	99.56±0.80	96.57±1.75	**100.0±0.0**
C15	93.07±0.24	93.88±0.53	93.26±1.89	91.94±6.58	76.44±38.3	99.70±0.37	95.62±3.25	94.29±1.89	**99.78±0.27**
C16	96.59±0.62	99.65±0.21	99.29±0.22	92.59±5.22	98.78±0.48	99.91±0.12	99.98±0.04	97.37±1.49	**100.0±0.0**
OA	96.59±0.62	97.00±0.26	95.66±0.52	88.82±0.25	92.38±5.07	95.52±6.07	98.55±0.90	97.31±0.23	**99.81±0.06**
AA	96.20±0.68	96.67±0.30	95.52±0.58	87.62±2.49	94.80±3.48	96.71±3.10	99.25±0.42	95.72±0.32	**99.90±0.04**
Kappa	98.64±0.27	98.86±0.12	98.29±0.17	94.97±1.63	91.55±5.76	95.07±6.61	98.38±1.01	96.99±0.26	**99.79±0.07**

TABLE VII

NUMBERS OF TRAIN AND TEST SAMPLES FOR THE HU DATASET

Class	Class name	Training	Test	Total
C1	Grass Healthy	80	1171	1251
C2	Grass Stressed	80	1174	1254
C3	GrassSynthetic	80	617	697
C4	Tree	80	1164	1244
C5	Soil	80	1162	1242
C6	Water	80	245	325
C7	Residential	80	1188	1268
C8	Commercial	80	1164	1244
C9	Road	80	1172	1252
C10	Highway	80	1147	1227
C11	Railway	80	1155	1235
C12	Parking Lot 1	80	1153	1233
C13	Parking Lot 2	80	389	469
C14	Tennis Court	80	348	428
C15	Running Track	80	580	660
Total		1200	13829	15029

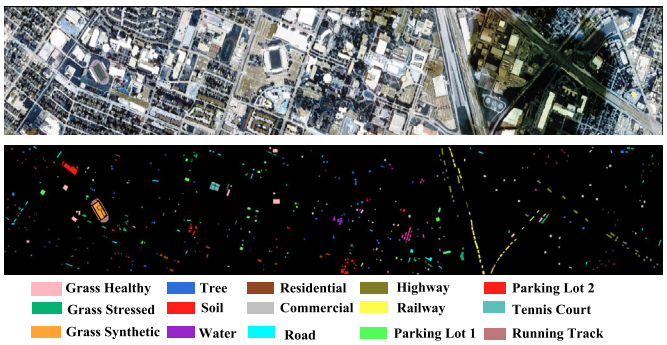


Fig. 14. False-color image and ground-truth map for the HU dataset.

We observed a significant reduction in edge error rate, with only a few edge pixels being misclassified, with good boundaries in targets such as railway, and effectively maintaining the clarity of edge information for each category in the HSI. Despite the poor performance in some hard-to-identify categories, it still achieves good classification accuracy due to its ability to capture a larger range of homogeneity and reduce misclassification. The pyramid structure is used in the framework, and the aggregation of multiscale information is

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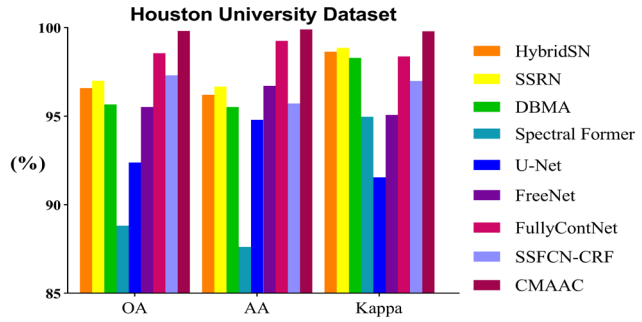


Fig. 15. Comparison of different experimental results on the HU dataset.

conducive to HSI classification. The best classification result in CMAAC proves the superiority of the model.

Table VIII lists the classification results on the HU dataset. CMAAC achieves the state-of-the-art results, outperforming HybridSN, SSRN, DBMA, SpectralFormer, FreeNet, FullyContNet, and SSFCN-CRF in terms of accuracy. Since the HU dataset contains some very challenging categories and complex spatial structures, DBMA, HybridSN, and SpectralFormer get relatively poor classification accuracies. FreeNet, FullyContNet, SSFCN-CRF, and our methods employ a global learning framework with superior classification accuracy on OA, AA, and Kappa than the patch-based methods. By replacing the patch with a larger receptive field, the patch-free global learning framework considers the global spectral–spatial information to improve the classification performance. Our model achieves the highest classification accuracy. Moreover, compared with FullyContNet, it increased OA, AA, and Kappa by 1.26%, 0.65%, and 1.41%, respectively. This improvement is due to the deeper network of the CMAAC, which can effectively utilize the spectral–spatial information of HSI and perform better transfer between convolutions. Notably, the PEAM mechanism retains more spatial information, enhances the representation capability of land-cover types at different scales, and makes high-level information more complete, significantly impacting complex data processing. Furthermore, the skip connections between the encoder and decoder in the shallow layers can refine semantically stronger features and improve the robustness of the model.

V. Discussion

To better demonstrate the effectiveness of each module in the CMAAC network, we conducted ablation experiments to analyze the contributions of all components on the IP, SA, PU, and HU datasets. We used the network without any strategy as the baseline, and the results are shown in Table IX, joint loss function abbreviated as JLF.

A. Discussion on the CLMS Model and PEAM Model

The effectiveness of the CLMS and PEAM modules has been described in Section III. Adding the CLMS module to the CMAAC baseline, the OA increases from 91.82% to 92.49% in the IP dataset. The accuracy of the CLSTM module alone is only 92.15%; the effective fusion of the two branches can capture multiorder interactions in context and extract discriminative spectral–spatial features to improve the model's classification performance. Adding the PEAM module to the CMAAC network increases the value of OA from 90.27% to 99.64%. This module is used for adaptive weighting to obtain multiscale spatial information, to resolve scale differences better, to complement the correlation of channels and spatial, and to improve the segmentation performance at different scales. As shown in Table IX, the PEAM module yields significant improvements, particularly on the SA and HU datasets, indicating that it can play a vital role in complex datasets. We can also see the variation in the accuracy of each mechanism in Fig. 17.

B. Discussion on the ACS Model and JLF Model

The ACS model enhances the influence of local key points features to complement the mesh convolution, which helps to recover detailed features of the target and improve the robustness of the model. Table IX shows the results of the ablation experiments performed by CMAAC. On the IP dataset with samples imbalanced, OA increases from 98.09% to 98.35%, AA from 99.09% to 99.17%, and Kappa from 97.83% to 98.12% after using the JLF. These quantitative results show that the loss function helps HSI to perform classification, especially for the imbalanced sample classification task. The model further improves the accuracy mainly by speeding up the model convergence through the weighted loss function, reducing the intraclass distance and increasing the interclass distance, and emphasizing the category prone to misclassification.

C. Discussion on the Running Time

The detailed training and test run time costs on FreeNet, FullyContNet, SSFCN-CRF, and CMAAC for each dataset are listed in Table X. The number of iterations for the comparison methods is set according to the optimal settings of the source literature. Notably, the inference time cost of all frameworks is substantially lower than their training time cost. The training time of CMAAC is 2–3 times longer than that of FreeNet, and the testing time is similar, but the classification accuracy has improved significantly. The training time is close to the FullyContNet, and the training time is lower than that of SSFCN-CRF, but the accuracy is slightly higher than FullyContNet and SSFCN-CRF. Based on the abovementioned analysis, under the condition of sufficient training time, the proposed CMAAC is worth to be the first choice. However, for scenarios with high timeliness requirements, we still need to further seek the balance between speed and accuracy.

D. Effect of Training Samples

The supervised information extracted during training is one of the critical factors affecting classification accuracy. To more fully compare the adaptability of different methods to various training sample sizes, we randomly selected {20, 50, 80, 110} labeled data from PU, IP, SA, and HU for training. The experimental results are shown in Fig. 18. As expected, the OA of all classification models is increased with the number of labeled samples grows. Constrained by

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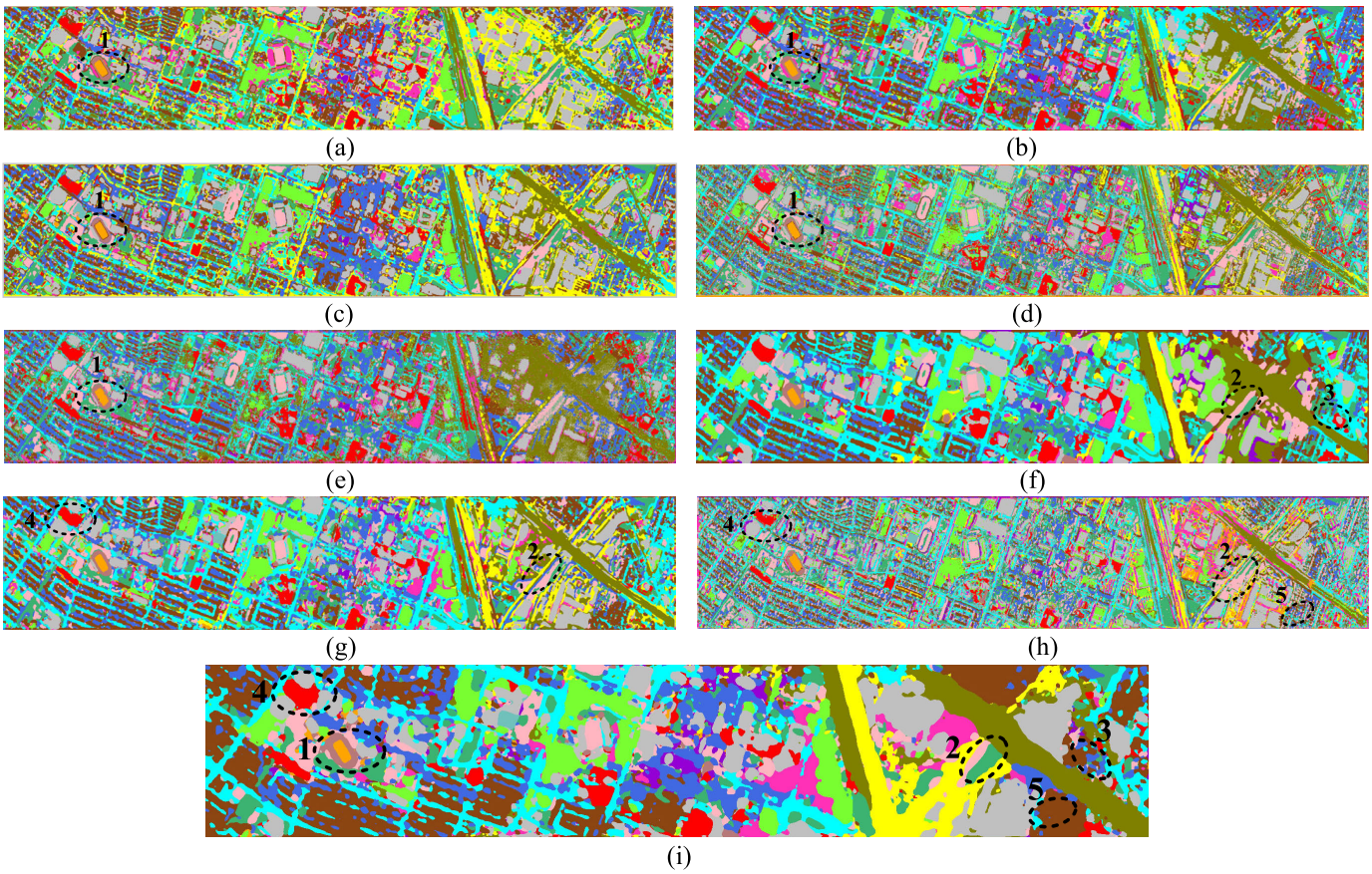


Fig. 16. Visualization of the classification maps for the HU dataset. The black dashed lines outline part of the subtle classification distinctions. (a) HybridSN. (b) SSRN. (c) DBMA. (d) SpectralFormer. (e) U-Net. (f) FreeNet. (g) FullyContNet. (h) SSFCN-CRF. (i) CMAAC.

TABLE VIII

CLASSIFICATION RESULTS OF HYBRIDSN [21], SSRN [42], DBMA [31], SPECTRALFORMER [52], U-NET [61], FREENET [34], FULLYCONTNET [35], SSFCN-CRF [23], AND CMAAC ON THE HU DATASET

	Patch-batch				Patch-free				
Class	HybridSN	SSRN	DBMA	SpectralFormer	U-Net	FreeNet	FullyContNet	SSFCN-CRF	CMAAC
C1	90.94±4.75	91.76±2.59	93.69±2.26	**99.86±0.04**	97.67±1.69	98.88±1.42	96.97±2.36	97.40±0.67	98.14±1.98
C2	97.86±1.17	96.35±1.46	97.85±1.89	97.10±2.22	91.88±4.0	94.67±6.02	**99.64±0.20**	93.31±1.33	98.87±1.06
C3	**99.94±0.07**	99.55±0.12	97.66±1.45	98.70±0.01	95.94±4.98	99.89±0.11	99.72±0.29	98.01±1.15	99.90±0.09
C4	97.56±1.43	95.36±0.89	96.51±0.84	91.71±0.13	98.04±1.78	97.01±2.61	**99.42±0.47**	94.42±3.033	98.26±1.72
C5	98.88±0.51	99.92±0.12	99.78±0.04	99.91±0.01	99.85±0.28	99.89±0.26	99.98±0.04	97.03±0.35	**100.0±0.0**
C6	99.41±0.83	97.63±1.66	98.08±1.52	95.10±2.04	97.29±1.37	98.58±3.67	99.59±0.71	93.27±4.29	**99.63±0.74**
C7	88.65±2.34	88.26±1.67	86.05±2.50	86.58±2.90	88.25±6.63	95.85±4.19	96.63±1.03	90.34±2.67	**98.64±1.08**
C8	85.74±3.38	79.63±5.85	82.20±6.68	88.44±0.30	76.09±5.07	78.76±7.79	**92.93±2.37**	87.03±2.01	91.01±1.41
C9	79.81±3.37	81.61±4.20	79.57±1.82	82.85±5.21	82.59±6.45	97.08±1.56	94.45±1.64	87.54±2.84	**97.13±1.56**
C10	89.91±3.57	97.66±1.74	94.74±4.05	93.77±2.92	90.64±8.38	84.59±30.1	**99.63±0.30**	95.40±0.37	99.51±0.70
C11	95.56±0.92	91.63±2.19	92.53±1.53	86.15±1.64	71.47±9.61	90.98±10.3	98.53±1.04	88.57±0.68	**99.11±1.41**
C12	89.63±4.36	96.15±2.30	87.11±3.96	91.46±2.39	79.85±7.45	98.24±1.47	96.81±1.27	92.28±1.56	**99.01±1.25**
C13	98.52±1.05	95.44±0.74	96.66±0.39	78.15±2.31	98.83±1.30	96.23±1.94	**99.94±0.11**	75.45±5.64	98.51±1.95
C14	99.91±0.12	99.66±0.47	99.83±0.24	100.0±0.0	99.98±0.07	**100.0±0.0**	**100.0±0.0**	97.06±2.52	**100.0±0.0**
C15	99.78±0.30	99.47±0.46	99.10±0.71	99.74±0.08	95.56±5.53	**100.0±0.0**	**100.0±0.0**	96.94±0.87	**100.0±0.0**
OA	91.80±0.82	93.02±0.28	90.97±0.88	92.23±1.08	89.26±1.59	94.56±2.90	97.78±0.38	92.43±0.91	**98.17±0.36**
AA	91.11±0.89	92.46±0.97	90.23±0.95	91.67±1.17	90.86±1.41	95.38±2.31	97.60±0.41	91.95±0.66	**98.44±0.32**
Kappa	93.26±0.71	93.97±0.58	92.26±0.59	92.64±0.84	88.39±1.72	94.12±3.13	98.22±0.28	91.81±0.99	**98.03±0.39**

the traditional CNN learning characteristics, a perfect classification model usually demands adequate support of labeled samples. More specifically, when there are only 50 training samples per category, the OA of CMAAC exceeds 95% on all four datasets. Even when there are only 20 training samples per category, the classification results are comparable

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TABLE IX
RESULT OF ABLATION EXPERIMENTS IN IP, SA, PU, AND HU DATASETS

Method	CLMS	PEAM	ACS	JLF	OA (%)		AA (%)		Kappa (%)	
					PU	IP	PU	IP	PU	IP
Baseline					96.73±1.98	91.82±1.42	97.92±0.75	80.03±3.02	95.73±2.55	90.71±1.61
Baseline	√				98.18±0.74	92.49±1.73	98.18±0.74	83.51±2.30	97.60±0.96	91.46±1.96
Baseline	√	√			98.75±0.45	97.90±0.43	98.75±0.14	98.77±0.24	98.36±0.59	97.61±0.48
Baseline	√	√	√		99.15±0.17	98.09±0.31	99.14±0.31	99.09±0.11	98.88±0.22	97.83±0.35
Baseline	√	√	√	√	**99.31±0.43**	**98.35±0.31**	**99.16±0.13**	**99.17±0.20**	**99.18±0.32**	**98.12±0.36**
					SA	HU	SA	HU	SA	HU
Baseline					88.67±8.90	83.72±2.11	94.49±3.60	85.90±1.80	87.57±9.70	82.41±2.28
Baseline	√				90.27±9.21	84.67±3.37	95.52±3.99	86.73±2.84	89.33±10.05	83.44±3.64
Baseline	√	√			99.64±0.16	96.36±0.64	99.86±0.05	96.92±0.57	99.60±0.18	96.08±0.69
Baseline	√	√	√		99.70±0.14	97.29±0.71	99.87±0.06	97.65±0.55	99.66±0.16	97.10±0.46
Baseline	√	√	√	√	**99.81±0.06**	**98.17±0.36**	**99.90±0.04**	**98.44±0.32**	**99.79±0.07**	**98.03±0.39**

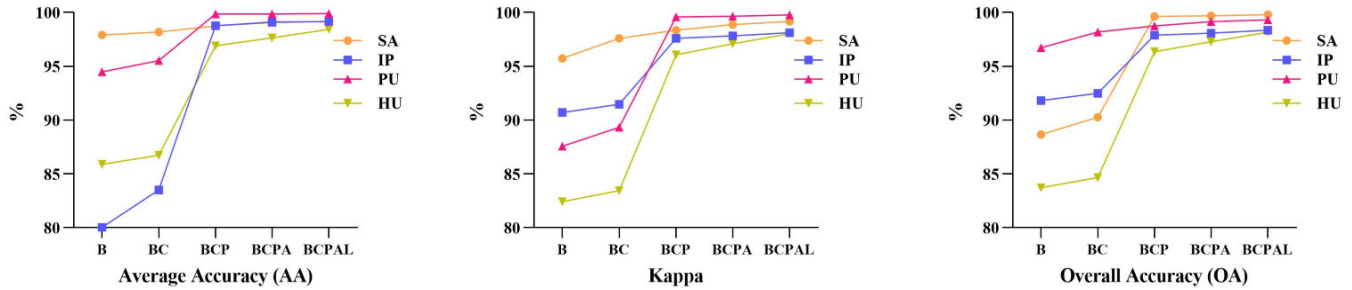


Fig. 17. Ablation experimental results on four datasets: PU, IP, SA, and HU.

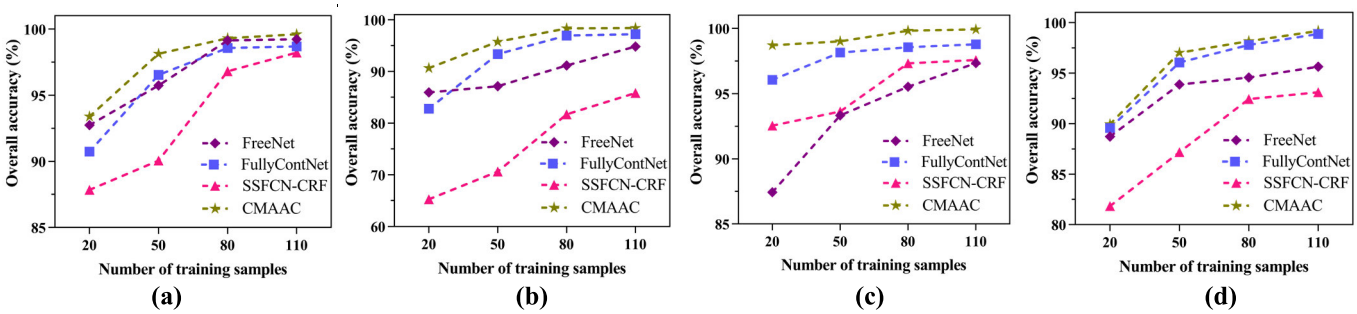


Fig. 18. Classification performance of different methods according to the variation of number of training samples. (a) PU. (b) IP. (c) SA. (d) HU.

TABLE X
TRAINING AND TESTING TIMES OF DIFFERENT METHODS ON FOUR HSI DATASETS

		IP	SA	PU	HU
FreeNet	Train	7.14	34.37	36.75	372.47
	(s)	±0.52	±1.24	±0.12	±3.63
	–Test	0.52	2.75	5.12	14.04
	(s)	±0.05	±0.03	±0.04	±0.08
FullyContNet	Train	61.64	87.41	92.13	730.54
	(s)	±0.42	±0.61	±0.72	±0.37
	Test	0.10	0.14	0.14	0.25
	(s)	±0.01	±0.01	±0.01	±0.04
SSFCN-CRF	Train	130.18	454.08	438.21	3440.8
	(s)	±1.59	±3.09	±2.00	±1.53
	Test	0.11	0.32	0.29	0.79
	(s)	±0.01	±0.01	±0.05	±0.02
CMAAC	Train	31.66	59.48	66.71	638.26
	(s)	±0.51	±0.70	±0.13	±7.40
	Test	0.51	2.60	5.07	19.11
	(s)	±0.01	±0.02	±0.03	±0.18

to or better than those of the advanced deep-learning methods.

E. Applicability of the Model

The effectiveness of the model was demonstrated in Section IV. To further demonstrate the applicability of the model, we conducted experiments on the Wuhan unmanned aerial vehicle (UAV)-borne hyperspectral image (WHU-Hi)-HongHu data. The WHU-Hi dataset is a benchmark dataset for precise crop classification based on UAV-borne hyperspectral imagery. The WHU-Hi-HongHu dataset was acquired from 16:23 to 17:37 on November 20, 2017, in Honghu city, Hubei province, China, with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro UAV platform. The experimental area is a complex agricultural scene with a wide variety of crops and different varieties of the same crop are also grown in the area, including Chinese cabbage and cabbages, *Brassica chinensis*, and small *B. chinensis*. The image size is 940 × 475 with 270 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne HSI is about 0.043 m. There are 355 638 labeled samples in the scene, which are classified into 22 different classes.

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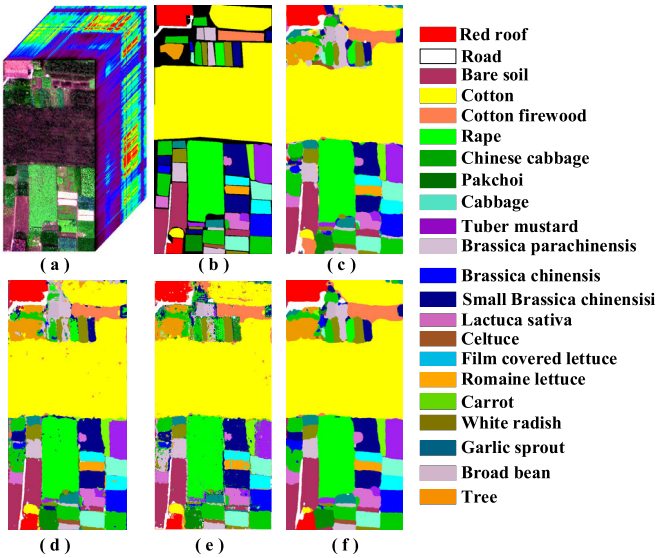


Fig. 19. Visualization of the classification maps for the WHU-Hi-HongHu dataset. (a) False-color image, (b) ground-truth map, (c) FreeNet (94.49%), (d) FullyContNet (96.30%), (e) SSFCN-CRF (93.30%), and (f) CMAAC (97.45%).

In the experiments, we select 50 samples of per class as a train set and the rest as a test set. Fig. 19(b) shows the ground-truth map, and Fig. 19(c)–(f) displays the classification result maps of different patch-free methods on the WHU-Hi-HongHu dataset. From the perspective of OA, the accuracy of CMAAC is 2.96%, 1.15%, and 4.15% higher than that of FreeNet, FullyContNet, and SSFCN-CRF, respectively. All the above models consider spectral–spatial information, which contributes to improve the accuracy. The FullyContNe further captures the nonlocal contextual information and acquires the dependencies between multiscale features, which also obtains better classification accuracy. However, our proposed CMAAC strengthens the long-term dependence of spectral features, captures multiorder interaction information, and enhances the ability to learn local key points. Fig. 19(f) is closer to the ground-truth map, which reflects the applicability of the model.

VI. Conclusion

In this article, we have proposed a new CMAAC global learning framework for end-to-end classification based on HSI. CMAAC was an FCN-based method designed to address the problem of different land covers with similar spectral information and capture the abundance of multiorder features, moreover, alleviates the loss of information in the flow of features, and extracts fine-grained multiscale spatial information. In the CMAAC framework, a CLMS mechanism consisting of CLSTM and MAGB is proposed to exploit the dependencies between spectral, complement the spectral–spatial correlation, and extract more discriminative features. The PEAM adaptively weights feature maps of different scales, channels, and spatial to extract fine-grained multiscale spectral–spatial information. In addition, the use of asymmetric convolution enhances the robustness of the framework, and the JLF enhances the feature distinction between different land covers, improving the convergence speed of the framework, which effectively alleviates the problem of imbalance between hard and well-classified samples. Each feature extraction module in the article can be integrated into different architectures for HSI classification. Experimental results on four widely used HSI datasets show that the proposed model achieves the best performance in terms of OA, AA, and Kappa and has a competitive advantage over the state-of-the-art methods. In future work, we will explore a more effective framework for HSI classification.

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